Global Mapper – Role & User Management (RBAC)

Summary

Introduce a scalable, extensible Role-Based Access Control (RBAC) framework within the Global Mapper tool that enables Admin users to create roles, manage permissions, and assign access to users with granular control across Mapping, Entities, and Customization modules.

Business Value

Global Mapper is a foundational platform layer impacting Trading, Product, Tech, Content, and Operations.

As the system grows in scope and complexity, granular access control becomes critical to:

- Ensure secure and controlled access to sensitive mapping and customization features
- Enable operational independence without hardcoded permission logic
- Support scalable growth of new GM features without re-engineering access controls
- Maintain auditability and compliance readiness
- Reduce dependency on engineering for access changes

This feature transforms Global Mapper from a controlled engineering tool into a scalable enterprise-grade platform component.

Success Metrics

- 100% feature access governed by RBAC
- No hardcoded permission checks
- Admins can manage access without engineering support
- Audit trail available for compliance

Problem Statement

Currently, Global Mapper does not have a flexible role and permission management framework.

Without a structured RBAC system:

- Feature access becomes static or environment-based
- Scaling new modules introduces security gaps
- Access governance lacks transparency

- Auditing changes to access is difficult
- Operational ownership is limited

Objectives

- Enable Admins to create and manage users
- Enable Admins to create and manage roles
- Provide granular permission control over GM features
- Ensure future extensibility for new features
- Maintain audit logs for all access changes

User Personas

1️⃣ Admin (Primary Persona)

- Responsible for access governance
- Creates users
- Creates and modifies roles
- Assigns permissions
- Reviews audit logs

2️⃣ Power User

- Advanced user managing entities and mappings
- May create/update mappings and customization
- Access governed by assigned role

3️⃣ Standard User

- Limited operational user
- Primarily view access
- Restricted modification capability

⚠️ Note: Users are **Brand-agnostic** as Global Mapper operates as a unified layer.

Major User Journeys

Journey 1: Create a New Role

1. Admin navigates to Role Management
2. Admin creates a new role

3. Admin assigns permissions (Entity, Mapping, Customization)
4. Role becomes available for assignment

Journey 2: Assign Role to User

1. Admin creates or edits a user
2. Admin selects one or more roles
3. User inherits permissions dynamically

Journey 3: Modify Role Permissions

1. Admin edits existing role
2. Adjusts feature-level access (View, Create, Update, Delete, etc.)
3. Changes reflect immediately for assigned users
4. Audit entry created

Journey 4: Governance & Audit

1. Admin reviews role changes
2. Views historical changes (who, what, when)
3. Ensures compliance traceability

Scope Overview

This initiative enables:

- Creation and management of Users
- Creation and management of Roles
- Assignment of one or more roles to users
- Granular permission control across GM modules
- Full audit logging of role and permission changes

Global Mapper operates as a **brand-agnostic layer**, therefore roles such as Power User or Standard User are not brand-specific.

Coverage

1️⃣ Role Management

- Create Role

- Update Role
- Delete Role (with guardrails)
- Clone Role
- Assign permissions per feature

2️⃣ User Management

- Create User
- Assign roles during creation
- Update assigned roles
- Activate / deactivate user

3️⃣ Permission Matrix Coverage

Entity Controls (CRUD separately)

- Sport
- Category
- Competition
- Event
- Markets
- Market Types (Templates)

Mapping Controls

- View Mapping
- Create Mapping
- Update Mapping
- Unmapping
- Pause Mapping
- Cascade above actions to Children

Customization Controls

- View Customization
- Create Customization
- Update Customization
- Delete Customization

Non-Functional Requirements

- Permission checks must not degrade system performance

- Access validation must be enforced consistently across UI
- No feature should be accessible without permission verification
- Audit logs must be immutable and traceable

Extensibility Requirement

[Note for Architecture and Design]

Any new GM feature introduced in the future must:

- Be added to the Role Permission Matrix
- Require explicit permission configuration
- Not default to open access

RBAC must scale alongside product growth.

Risks & Considerations

- Role sprawl without governance (***Already faced in the Current PTC***)
- Accidental privilege escalation
- Complex cascading permission logic
- Improper deletion of active roles

Dependencies

- Global Mapper UI
- Mapping Module
- Customization Module
- Entity Management screens
- Authentication system (SSO)

Q&A

Area	Query	Answer